Supplementary Material for "Certified Federated Unlearning for IoT: Ledger-Anchored Revocation with Anytime-Valid Forgetting Certificates"

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I. Setup and cost model

All experiments are produced by the scripts in the repository (`experiments/unlearning_experiments.py`, `unlearning_real.py`, `unlearning_new.py`); tables below are written from the result files by `paper/make_supp_tables.py`.

Training. Binary logistic loss, full-batch local gradient, update clipping at norm 1, learning rate 1, $T = 60$ rounds, participation $p = 0.6$ (independent per client and round), reputation rate 0.1 and penalty 0.2, Pedersen commitments with 512-bit group, spectral split gated by a separation margin of 0.25, no differential-privacy noise unless stated. Synthetic fleet: 60 clients, $d = 1024$, 12 real records and 6 canary pairs per client, Dirichlet(α=1) over ten latent classes, 8% label noise. UCI HAR: 30 subjects, 40 records each. N-BaIoT: nine devices, five sites each, 40 records each, features $\mathrm{sign}(x)\log(1+|x|)$ standardised. The MLP has 32 tanh hidden units. Test records are held out per client. Audits use $\epsilon = 0.2$, $\alpha = 0.05$ and a reveal order drawn from a fixed seed.

Cost model. A client round with n records costs $n \cdot f$ FLOPs ($f = 4d$ for the linear model, $6P$ for an MLP with P parameters), uploads $4P + 64$ bytes (float32 update and a 512-bit commitment) and downloads $4P$ bytes. Energy is $e_{\mathrm{flop}}\cdot\mathrm{FLOPs}+e_{\mathrm{bit}}\cdot 8\cdot\mathrm{bytes}$ with $e_{\mathrm{flop}} = 1$ nJ and $e_{\mathrm{bit}} = 100$ nJ; these are assumed constants, not measurements. With them radio traffic dominates, so energy follows bytes. Canary reveal sends $2m(d{+}1)\cdot 4$ bytes per revoked client once.

II. Additional figures

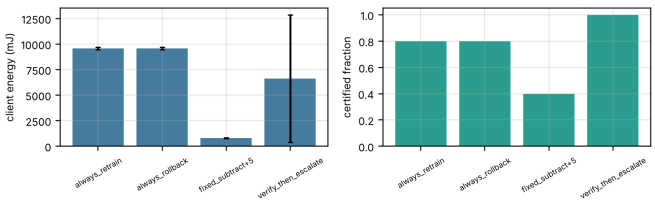


Fig. S1. Synthetic fleet, wave of 20 clients (10 seeds): mean client energy (bars, s.d. as error bars) and share of waves certified for the verify-then-escalate ladder, always retraining, always rollback and a fixed subtract+5 policy.

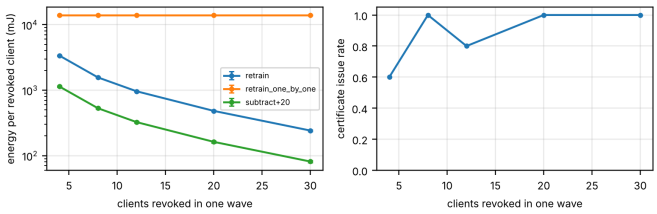


Fig. S2. Synthetic fleet: energy per revoked client versus wave size (left; "one by one" is measured on three requests and extrapolated) and issue rate of the certificate for the retrained model (right, 5 seeds).

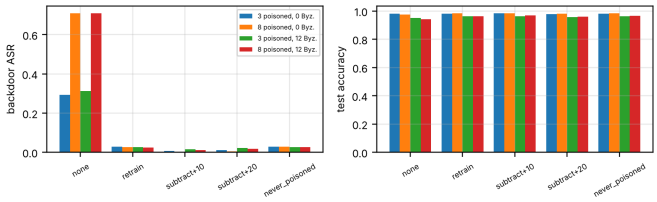


Fig. S3. Synthetic fleet, backdoor success rate and test accuracy before and after unlearning, with 3 or 8 poisoned clients in the wave and with 0 or 12 Byzantine sign-flipping clients (5 seeds).

III. Additional tables

IV. Reproduction

```
pip install -r requirements.txt
python experiments/unlearning_experiments.py    # s
python experiments/unlearning_real.py --dataset ha
python experiments/prepare_nbaiot.py            # o
python experiments/unlearning_real.py --dataset nb
python experiments/unlearning_new.py            # I
python experiments/fig_new.py
python paper/make_supp_tables.py
python tests/test_unlearning.py
```

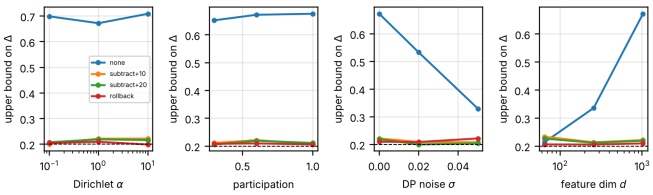


Fig. S4. Synthetic fleet: upper bound on the residual advantage against the Dirichlet parameter, participation rate, DP noise and feature dimension for the un-unlearned model and three operators (5 seeds). Dashed: $\epsilon = 0.2$.

TABLE S1
SYNTHETIC FLEET, ONE WAVE OF 20 OF 60 CLIENTS, 10 SEEDS. I/U/R: CERTIFICATE ISSUED, UNDETERMINED, REVOKED. Δ_{up}: MEAN UPPER BOUND ON THE RESIDUAL ADVANTAGE.

Operator	I/U/R	Acc.	Δ_{up}	Rounds	MB	Energy (J)
none	0/0/10	0.981	0.667	0	0.00	0.00
subtract	4/1/5	0.620	0.243	0	0.00	0.00
replay	4/1/5	0.621	0.252	0	0.00	0.00
subtract+5	4/1/5	0.983	0.249	5	1.00	0.81
subtract+10	4/3/3	0.982	0.226	10	2.03	1.64
subtract+20	6/3/1	0.981	0.212	20	4.00	3.24
rollback	8/2/0	0.983	0.212	60	11.84	9.58
retrain	8/2/0	0.983	0.212	60	11.84	9.58

TABLE S2
CALIBRATION: THE UN-UNLEARNED MODEL (SHOULD BE FLAGGED) AND THE EXACTLY RETRAINED MODEL (SHOULD NOT).

Data	Model	Model state	Runs	I/U/R
synthetic	linear	none	40	0/0/40
synthetic	linear	retrain	40	40/0/0
HAR	linear	none	60	0/0/60
HAR	linear	retrain	60	49/11/0
HAR	mlp	none	60	0/0/60
HAR	mlp	retrain	60	51/7/2
N-BaIoT	linear	none	60	42/9/9
N-BaIoT	linear	retrain	60	55/3/2
N-BaIoT	mlp	none	60	14/11/35
N-BaIoT	mlp	retrain	60	54/6/0

TABLE S3
REAL DATA, A WAVE OF ONE THIRD OF THE CLIENTS, 5 SEEDS.

Data	Model	Operator	I/U/R	Acc.	Rounds	Energy (J)
HAR	linear	none	0/0/5	0.999	0	0.00
HAR	linear	subtract	2/2/1	0.991	0	0.00
HAR	linear	replay	2/1/2	0.991	0	0.00
HAR	linear	subtract+5	2/2/1	0.998	5	0.22
HAR	linear	subtract+10	2/2/1	0.998	10	0.46
HAR	linear	subtract+20	3/1/1	0.998	20	0.91
HAR	linear	retrain	3/2/0	0.998	60	2.69
HAR	mlp	none	0/0/5	0.999	0	0.00
HAR	mlp	subtract	3/2/0	0.995	0	0.00
HAR	mlp	replay	3/2/0	0.995	0	0.00
HAR	mlp	subtract+5	4/1/0	0.998	5	7.21
HAR	mlp	subtract+10	4/1/0	0.998	10	14.90
HAR	mlp	subtract+20	4/1/0	0.998	20	29.24
HAR	mlp	retrain	4/1/0	0.999	60	86.48
N-BaIoT	linear	none	3/1/1	0.996	0	0.00
N-BaIoT	linear	subtract	4/1/0	0.871	0	0.00
N-BaIoT	linear	replay	4/1/0	0.942	0	0.00
N-BaIoT	linear	subtract+5	4/1/0	0.960	5	0.08
N-BaIoT	linear	subtract+10	4/1/0	0.995	10	0.15
N-BaIoT	linear	subtract+20	4/1/0	0.996	20	0.30
N-BaIoT	linear	retrain	4/1/0	0.995	60	0.88
N-BaIoT	mlp	none	0/0/5	0.997	0	0.00
N-BaIoT	mlp	subtract	5/0/0	0.962	0	0.00
N-BaIoT	mlp	replay	4/1/0	0.939	0	0.00
N-BaIoT	mlp	subtract+5	5/0/0	0.996	5	2.38
N-BaIoT	mlp	subtract+10	5/0/0	0.997	10	4.73
N-BaIoT	mlp	subtract+20	5/0/0	0.997	20	9.35
N-BaIoT	mlp	retrain	5/0/0	0.997	60	27.36

TABLE S4
RECOVERY POLICIES (5 SEEDS). "U" COUNTS UNDETERMINED AND UNPOWERED. LAST COLUMN: WORST RUN.

Data	Model	Policy	I/U/R	Acc.	Rounds	Mean J	Max J
Synthetic	linear	retrain	5/0/0	0.986	60.0	9.64	9.88
Synthetic	linear	ladder	5/0/0	0.985	41.0	6.68	13.81
Synthetic	linear	icr_q1.0	5/0/0	0.982	17.0	2.74	4.83
Synthetic	linear	icr_q0.5	4/1/0	0.982	16.0	1.30	4.81
Synthetic	linear	icr_q0.25	4/1/0	0.988	26.0	1.03	2.37
HAR	linear	retrain	3/2/0	0.999	60.0	2.69	2.74
HAR	linear	ladder	3/2/0	0.996	39.0	1.74	3.82
HAR	linear	icr_q1.0	4/1/0	0.996	24.0	1.07	2.63
HAR	linear	icr_q0.5	4/1/0	0.996	18.0	0.40	1.29
HAR	linear	icr_q0.25	3/2/0	0.996	27.0	0.34	0.75
HAR	mlp	retrain	4/1/0	0.999	60.0	86.48	88.12
HAR	mlp	ladder	4/1/0	0.997	18.0	25.56	119.58
HAR	mlp	icr_q1.0	4/1/0	0.997	13.0	18.54	84.50
HAR	mlp	icr_q0.5	4/1/0	0.997	13.0	9.02	41.47
HAR	mlp	icr_q0.25	4/1/0	0.997	13.0	5.09	23.63
N-BaIoT	linear	retrain	4/1/0	0.995	60.0	0.88	0.89
N-BaIoT	linear	ladder	1/4/0	0.980	24.0	0.36	1.25
N-BaIoT	linear	icr_q1.0	1/4/0	0.980	16.0	0.24	0.90
N-BaIoT	linear	icr_q0.5	1/4/0	0.981	15.0	0.11	0.45
N-BaIoT	linear	icr_q0.25	1/4/0	0.980	10.0	0.04	0.14
N-BaIoT	mlp	retrain	5/0/0	0.997	60.0	27.36	27.74
N-BaIoT	mlp	ladder	5/0/0	0.982	2.0	0.88	2.38
N-BaIoT	mlp	icr_q1.0	5/0/0	0.982	2.0	0.88	2.38
N-BaIoT	mlp	icr_q0.5	5/0/0	0.982	2.0	0.42	1.14
N-BaIoT	mlp	icr_q0.25	5/0/0	0.982	2.0	0.23	0.61

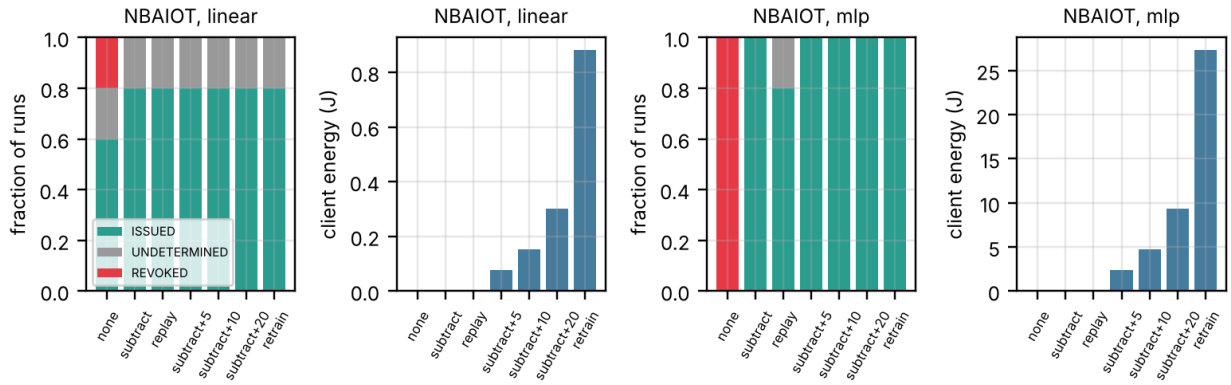


Fig. S5. N-BaIoT (45 clients, wave of 15, 5 seeds): certificate outcomes and client energy per operator for the linear model and the MLP.

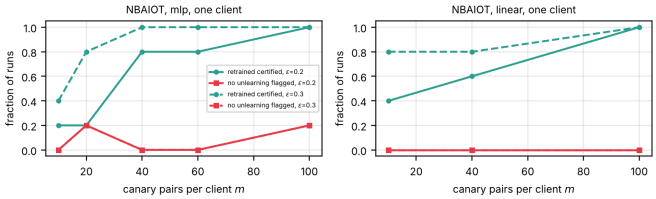


Fig. S6. N-BaIoT, one revoked client: share of runs in which the retrained model is certified and the un-unlearned model is flagged, against canary pairs per client. The audit has little power at these model sizes.

TABLE S5
CANARY DESIGN GRID: SHARE OF RUNS (5 SEEDS) IN WHICH THE AUDIT FLAGS THE UN-UNLEARNED MODEL.

Data	Model	m	Scale	Flagged	Δ_{up}	Acc.
N-BaIoT	linear	6	1	0.2	0.25	0.996
N-BaIoT	linear	12	1	0.6	0.22	0.996
N-BaIoT	linear	6	2	0.2	0.26	0.996
N-BaIoT	linear	12	2	0.6	0.21	0.967
N-BaIoT	linear	6	4	0.4	0.24	0.995
N-BaIoT	linear	12	4	0.6	0.21	0.932
N-BaIoT	linear	6	8	0.2	0.23	0.995
N-BaIoT	linear	12	8	0.8	0.22	0.967
N-BaIoT	mlp	6	1	1.0	0.36	0.997
N-BaIoT	mlp	12	1	1.0	0.35	0.997
N-BaIoT	mlp	6	2	1.0	0.50	0.997
N-BaIoT	mlp	12	2	1.0	0.52	0.997
N-BaIoT	mlp	6	4	1.0	0.57	0.997
N-BaIoT	mlp	12	4	1.0	0.59	0.997
N-BaIoT	mlp	6	8	1.0	0.46	0.997
N-BaIoT	mlp	12	8	1.0	0.56	0.997
HAR	linear	6	1	1.0	0.83	0.999
HAR	linear	12	1	1.0	0.65	0.999
HAR	linear	6	2	1.0	0.81	0.999
HAR	linear	12	2	1.0	0.61	0.999
HAR	linear	6	4	1.0	0.72	0.999
HAR	linear	12	4	1.0	0.61	0.999
HAR	linear	6	8	1.0	0.60	0.999
HAR	linear	12	8	1.0	0.56	0.999

TABLE S6
WAVE SCHEDULER SIMULATION, ONE YEAR, 20 RUNS, DEADLINE 30 DAYS, POWER TARGET 120 PAIRS.

Req./day	Policy	Requests	Waves	Mean lat.	Max lat.	J/request	Underpowered
0.2	immediate	73	66.2	0.5	1.0	2.48	1.00
0.2	batched	73	11.1	17.2	29.9	0.42	1.00
1	immediate	360	230.8	0.5	1.0	1.76	1.00
1	batched	360	18.1	10.3	29.6	0.14	0.07
4	immediate	1458	358.7	0.5	1.0	0.67	1.00
4	batched	1458	66.8	2.8	27.5	0.13	0.01

TABLE S7
BASELINES ON THE MLP (5 SEEDS): MARKER SUCCESS RATE, LOSS-THRESHOLD MEMBERSHIP AUC AND OUR CERTIFICATE OUTCOME.

Data	Model state	Marker ASR	MIA AUC	Canary I/U/R
HAR	none	0.034	0.438	0/0/5
HAR	retrain	0.034	0.438	4/1/0
HAR	subtract+20	0.030	0.438	4/1/0
N-BaIoT	none	0.395	0.426	0/0/5
N-BaIoT	retrain	0.432	0.416	5/0/0
N-BaIoT	subtract+20	0.442	0.414	5/0/0

TABLE S8
PROVENANCE LEDGER ON THE SYNTHETIC FLEET (3 SEEDS, 200 TAMPER TRIALS EACH).

Quantity	Value
records (60 rounds)	61
ledger size (kB)	101
bytes per round	1652
ledger / update traffic	0.011
chain verification (ms)	1.6
Merkle proof check (μs)	4.8
proof length (hashes)	5.7
tampering detected	1.00
checkpoints (kB)	417
stored updates (kB)	15154